

Claude Co-work for Academics

BUILDING AN AUTONOMOUS RESEARCH PARTNER

Adapted from Andy Stapleton's Claude Co-work Guide



Celebrating Aldrin! 🏳️‍🌈 ✨

Happy Pride! Celebrating our gay icon!

The Claude Model Family

FROM LIGHTWEIGHT TO ELITE REASONING

Claude Haiku

Ultra-fast, cost-efficient. Scanning emails, sorting library documents, classification routing.

Claude Sonnet

Balanced all-rounder. Frontier-level coding, interactive analysis, daily writing tasks.

Claude Fable

Narrative generation. Crafting research narratives, grant writing, and communicating complex findings.

Claude Opus

Elite deep reasoning. Dense PDF analysis, complex scientific synthesis, multi-stage workflows.

Claude Mythos

Exploratory reasoning. Ideal for interdisciplinary brainstorming and hypothesis generation.

Co-work leverages **Opus-class** deep cognition to power autonomous multi-agent background execution.

* Bai, Y., Jones, A., Chen, A., et al. (2022). "Constitutional AI: Harmlessness from AI Feedback." *arXiv preprint arXiv:2212.08073*.

The Shift: Chat vs. Co-work

THE AGENTIC PARADIGM

Traditional Chat

- **Linear ping-pong** — Requires constant prompting.
- **Amnesia** — Forgets previous papers or context.
- **Isolated** — Cannot run code or access local PDFs.
- **Manual** — You must copy-paste everything.

Claude Co-work

- **Objective-driven** — Runs autonomously in the background.
- **Persistent Memory** — Recalls your entire thesis structure.
- **Integrated** — Interacts directly with your file system.
- **Agentic** — Executes Python scripts, downloads papers.

Tasks take 10-15 minutes because the system is running multi-step code execution, searching external databases, and self-correcting.

Custom "Skills"

✨ Let's knock something off your list

REPEATABLE ACADEMIC WORKFLOWS

Standardize your complex prompts as saved, reusable execution files.

- **Saved Skills** — Package multi-step instructions (journal formatting, citation schemas) into single-click triggers
- **Backslash Trigger** — Type `\` in any workspace to list and instantly run your saved skills
- **Skill Creator Meta-Skill** — Capture any successful Claude conversation and save it as a permanent, reusable skill template

This guarantees execution consistency regardless of underlying model updates.

How can I help you today?

+

↑

📁 Work in a project ▾

Ask ▾

Sonnet 4.6 ▾

Connecting to Science: MCP Connectors

LIVE ACADEMIC DATABASE INTEGRATION

The **Model Context Protocol (MCP)** lets Claude directly query live external databases — resolving the hallucination problem at its root.

Consensus

Queries 200M+ peer-reviewed papers. Returns real abstracts, DOIs, and hyperlinked citations.

PubMed & bioRxiv

Direct integration with clinical studies, biology research, and preprints.

PubChem

Retrieves accurate chemical structures, compound profiles, and safety data.

Bio Render

Interfaces with scientific illustration tools for schematic diagrams.

Zero copy-pasting. Abstracts, metrics, and citation hyperlinks are pulled natively into the AI environment.

* Model Context Protocol (MCP) introduced by Anthropic (Nov 2024).

* Consensus database comprises >220 million peer-reviewed papers via Semantic Scholar & OpenAlex (2026).

Projects & Schedules

AUTOMATION WORKFLOWS

Project Sandboxes

- Upload style sheets, grant requirements, or journal layouts
- Toggle **Memory** to carry over context, definitions, and experimental data across chats
- Set project-level instructions like "Always write in IEEE format"

Recurring Schedules

- Automate repetitive literature searches on a weekly basis
- Schedule tasks like "Every Monday at 9 AM, query Consensus for new OPV papers and compile a summary report"
- Enable **Keep computer awake** for weekend background jobs

Monday 9:00 AM



Query Consensus



Filter by Impact



Summarize Findings



Save to Project Memory

Deep Dive: The 14-Page Literature Review

Thesis Lifecycle — Claude as Co-worker

INTERACTIVE



Lit Review



Methodology



Analysis



Drafting



Polishing

PHASE 1: LITERATURE REVIEW

Synthesizing across dozens of studies

Claude summarizes papers, extracts methods, and builds thematic comparison matrices from uploaded PDFs.

#Summarization

#Thematic Matrix

#Gap Analysis

EXAMPLE PROMPT

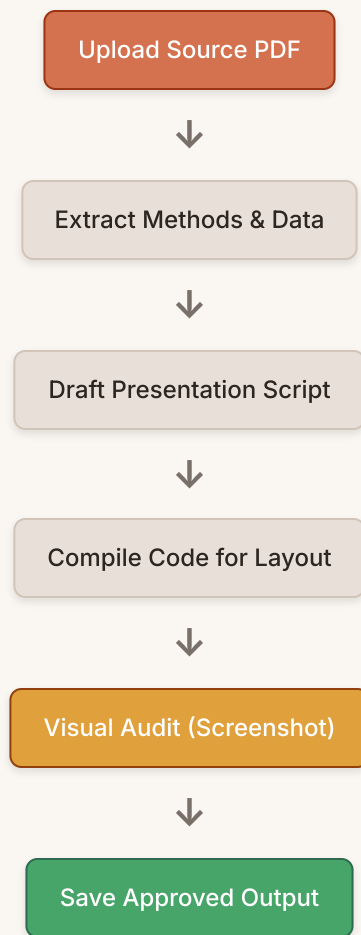
Upload 3 PDFs on AI in education. Extract research questions, methods, findings, and gaps. Format as a comparison table.

Deep Dive: PDF-to-Slides

MULTI-AGENT PRESENTATION PIPELINE

Convert dense scientific papers into formatted presentation structures.

- **Extraction** — Distills core arguments, methods, and data from source PDFs
- **Script Drafting** — Generates spoken scripts calibrated to target duration
- **Layout Compilation** — Runs local code execution to build slide templates
- **Visual Auditing** — Screenshot-based feedback loop detects and corrects overlapping text or layout issues



Settings for Agent Autonomy

ENABLE THESE IN FILE → SETTINGS → CAPABILITIES

To unlock Claude Co-work's full academic potential, toggle these settings **ON** in the desktop app:

Capabilities

- ✓ **History Memory** — Learns research styles & preferences.
- ✓ **Tool Access** — Grants local read/write file permissions.
- ✓ **Agent Vision** — Screenshot validation for auto-correction.
- ✓ **Code Execution** — Run R/Python statistical scripts locally.
- ✓ **Keep Awake** — Essential for background weekend runs.

Safeguards

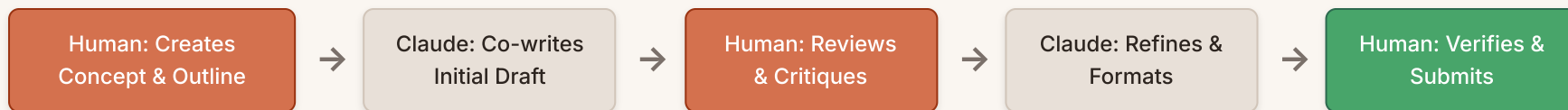
- ✓ **Require Approval** — Pauses before paying API costs.
- ✓ **File Restrictions** — Limit access to specific directories.
- ✓ **Budget Limits** — Hard caps on token usage per session.
- ✓ **Review Logs** — Inspect every bash command and script.

Ethical Boundaries

OWNERSHIP & ACADEMIC INTEGRITY

Claude is a tool, not an author. Maintain clear ethical boundaries.

- **Human Agency** — You are fully responsible for thesis content. Claude assists, but critical thinking must be yours.
- **Plagiarism vs. Editing** — Use Claude to improve phrasing, check grammar, or translate. Do not let it write original arguments unsupervised.
- **Citation Verification** — Claude can hallucinate references. Always verify every citation against Google Scholar, Scopus, or the original database.



Thank You

QUESTIONS & DISCUSSION

Let the AI handle the operational logistics, so you can focus on the science.